Curating Gene Alias Collisions for the Resolution of Gene Symbol Ambiguity

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Alias-Primary Collisions

HGNC ID	Gene Symbol	Aliases
7989	NRAS	NS6, CMNS, KRAS, N-ras, NCMS, NRAS1, ALPS4
6407	KRAS	NS, NS3, OES, CFC2, RALD, K-Ras, KRAS1…

Figure 1. Alias-Primary Collisions describe a gene symbol being used as a primary gene symbol and an alias, or synonym, for multiple genes. KRAS is used as a primary gene symbol for *KRAS* (hgnc:6407) and an alias for *NRAS* (hgnc:7989).

Alias-Alias Collisions

HGNC ID	Gene Symbol	Aliases
19048	ASPM	MCPH5, ASP
1318	C3	C3a, ASP
24059	TMPRSS11D	HAT, ASP

Figure 2. Alias- Alias Collisions describe a gene symbol being used as an alias for multiple distinct genes. The gene symbol ASP used as an alias for the genes *ASPM* (hgnc:19048), *C3* (hgnc:1318), *TMPRSS11D* (hgnc:24059).

Collision Occurence

Source	Total # of gene records	% of records with alias-primary collisions	% of records with alias-alias collisions
Ensembl	40,353	0.5% (218)	9.1% (3,680)
HGNC	43,164	1.1% (483)	4.8% (2,084)
NCBI Gene Info	75,346	2.27% (1,712)	7.53% (5,670)

Figure 3. Frequency of collisions in three databases.

We are developing a **framework** & **curation platform** to support **disambiguation** of shared gene symbols

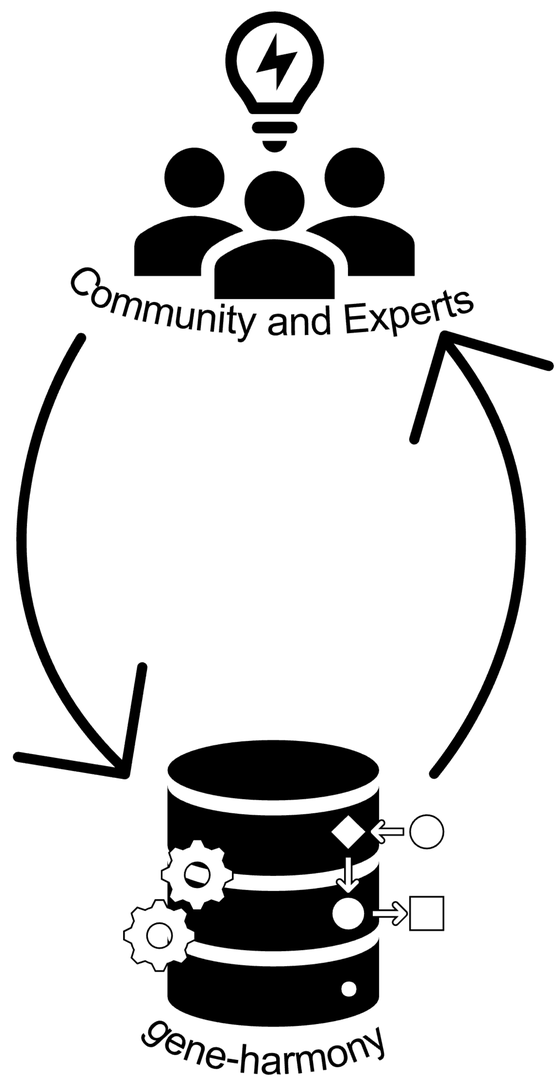

Figure 4. Incremental progress in annotating collisions promotes sustainable improvement and flexibility with community involvement.

gene-harmony SCAN ME

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Using Gene-Harmony to Resolve Collisions

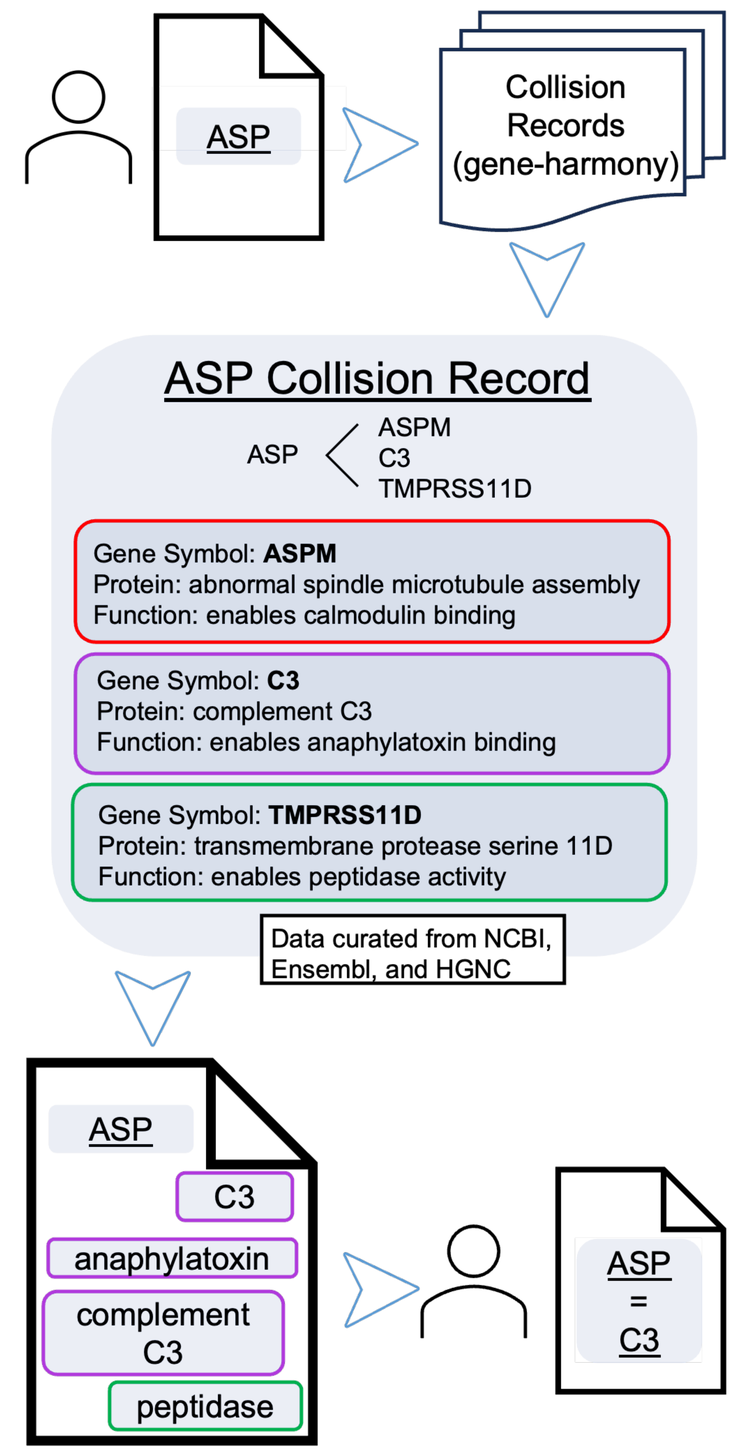

Figure 5. To resolve a collision, a user can refer to the collision curation platform which provides the framework for disambiguating shared symbols using the context of the document and the annotated collision record. This example shows a document with the ambiguous symbol ASP matching to the data from the collision associated gene C3, shown outlined in purple.

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